From Bed Rest to Mars

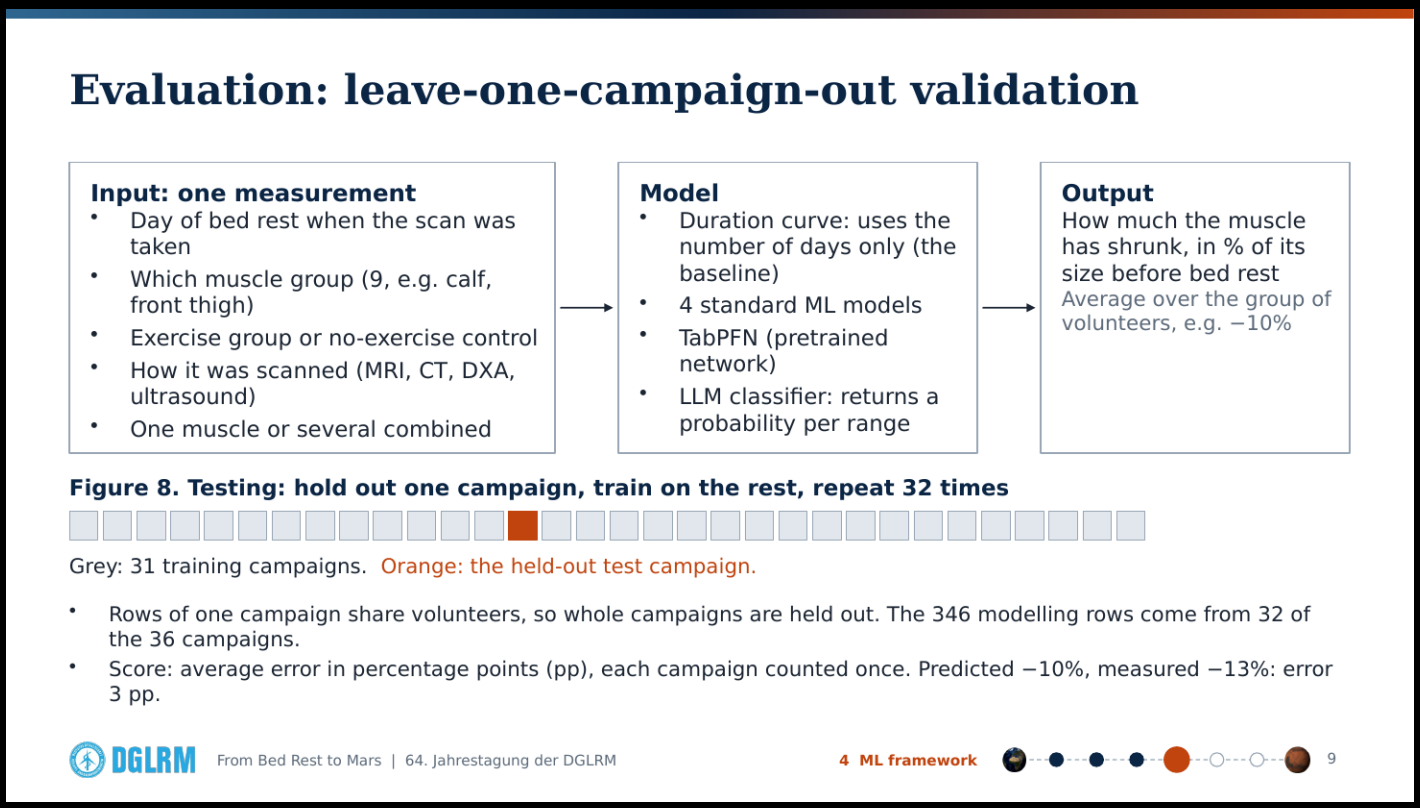
Practice script, Milad's part

Slides 9 to 14: about six minutes. Backup slides 15 to 21 are for questions only.

Amber italic lines are only for answering questions; do not read them out in the talk.

Slide 9 · Evaluation: leave-one-campaign-out validation

About one minute



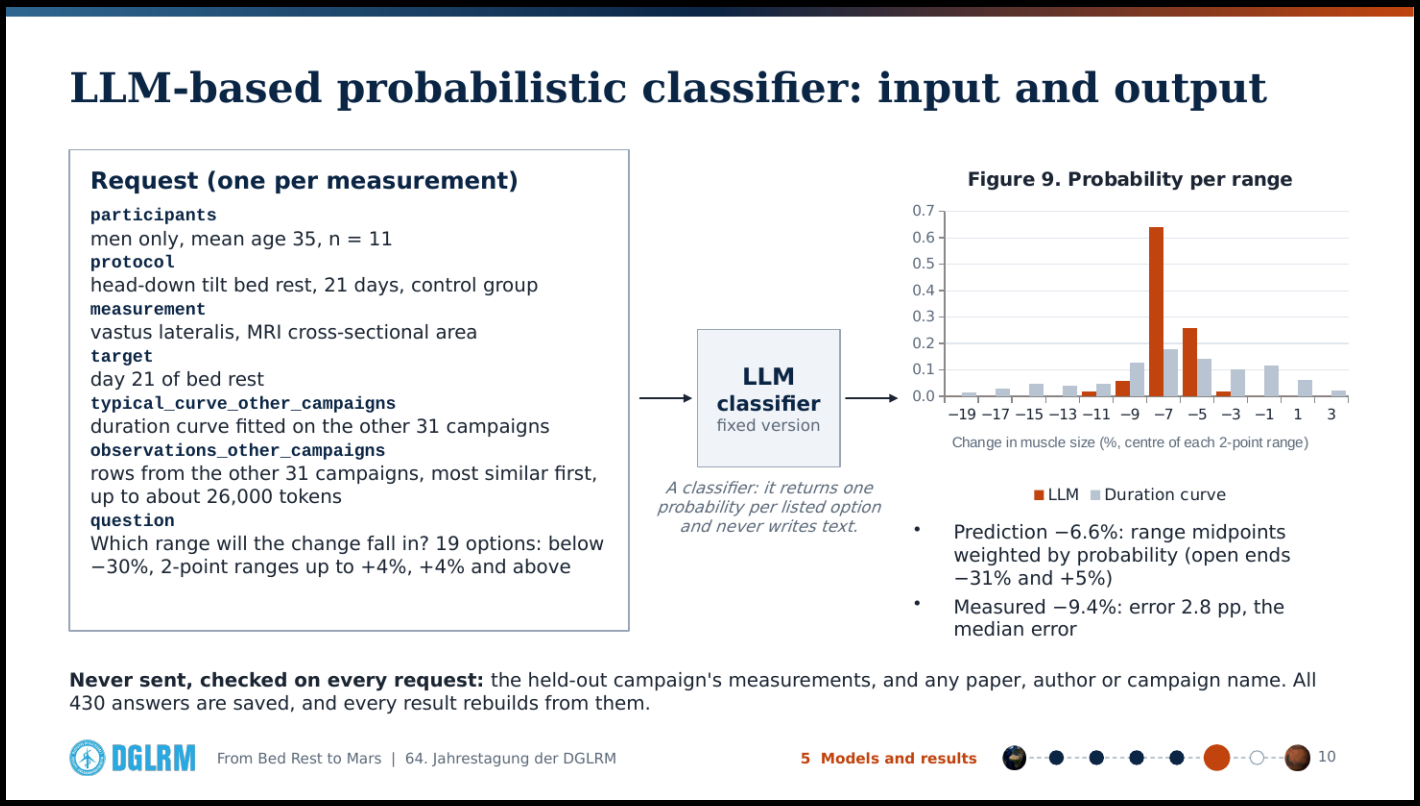
Each row in our data is one measurement: a group of volunteers, one muscle, one day of bed rest. The inputs are the day of the scan, the muscle family, whether the group did a countermeasure, how the muscle was imaged, and whether the value covers a group of muscles. The output is the percentage change in muscle size from before bed rest.

We have 346 rows, but they come from 32 independent campaigns, because several papers report the same volunteers. So we test by holding out one whole campaign, training on the other 31 and predicting the held-out one, 32 times. The code checks that no campaign appears on both sides of a split.

The score is the mean absolute error in percentage points, averaged so that each campaign counts once. Every model is compared with a duration-only curve, which uses the number of days and nothing else.

Slide 10 · LLM-based probabilistic classifier: input and output

About a minute and a half



The ML models only saw coded columns. Here we gave each measurement, written out in words, to a newly released language model built for probabilistic classification. It is not a chat model that writes text: you give it a described situation and a fixed list of options, and it returns a probability for each option, nothing else.

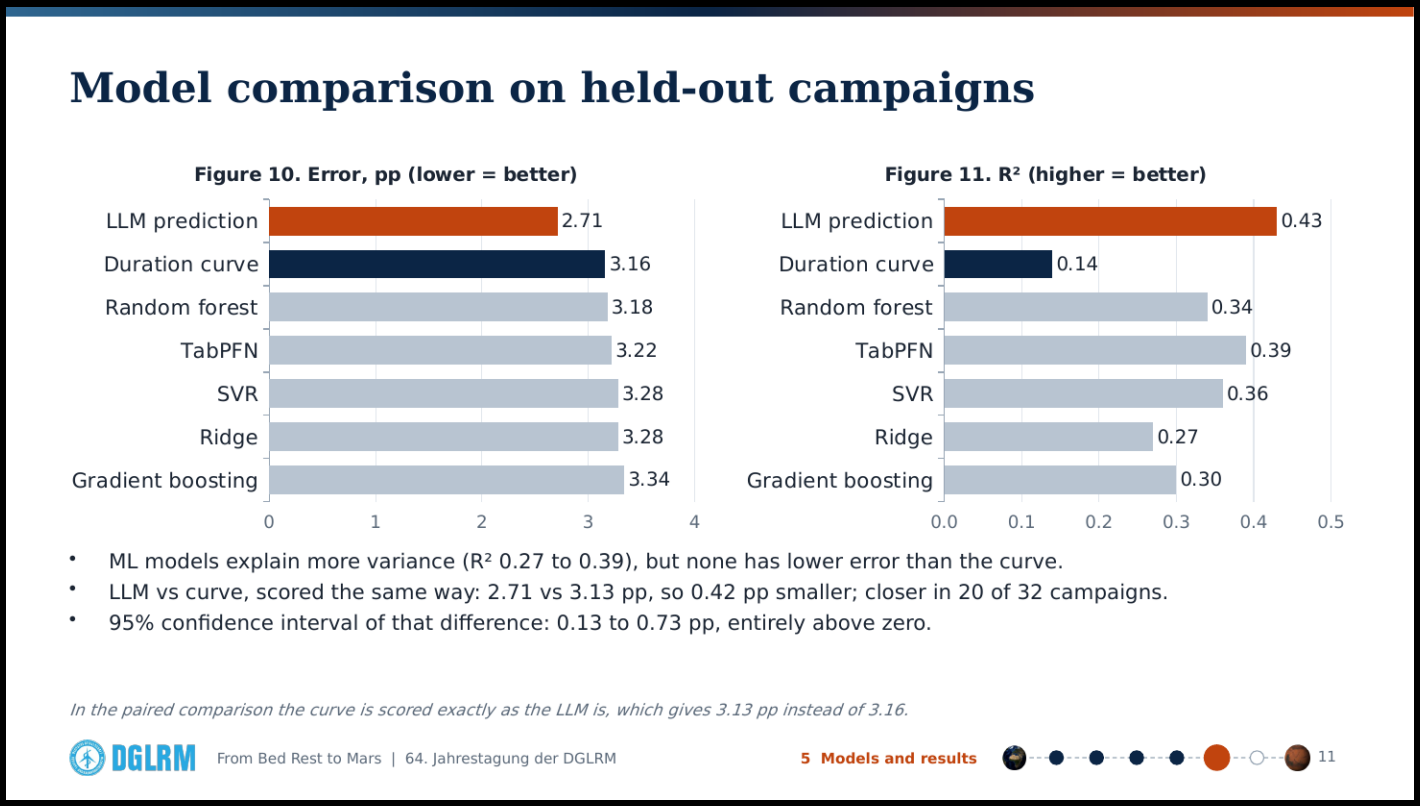
On the left is a real request from our cache. It describes the volunteers, the protocol, the muscle and how it was scanned, and the day. It also contains the duration curve fitted on the other 31 campaigns and as many rows from those campaigns as fit, most similar first. The question asks which of 19 ranges the change will fall in.

On the right is what came back for this row: a probability for each range. We never ask the model for a number. Our code turns the probabilities into a prediction, here minus 6.6 percent against an observed minus 9.4. We chose this row because its error is the model's median error.

The request never contains the held-out campaign's own data or any name that could identify a paper, and the code checks that on every request. Every answer is cached, so all our numbers can be rebuilt without calling the model again.

Slide 11 · Model comparison on held-out campaigns

About a minute and a half



We compared four standard models, ridge regression, random forest, support vector regression and gradient boosting, tuned inside each training fold, and later TabPFN, a neural network pretrained for small tables. On the left is the error on the held-out campaign. The duration curve scores 3.16 percentage points and the best ML model, the random forest, 3.18: none of them is lower than the curve. On the right, the ML models explain two to three times more variance, because they learn that the calf loses more than the hip. Most of the error left is a shift that applies to a whole campaign, from its scanner, its volunteers or its protocol, and none of our columns describes it. We had committed in advance to report this result whichever way it came out.

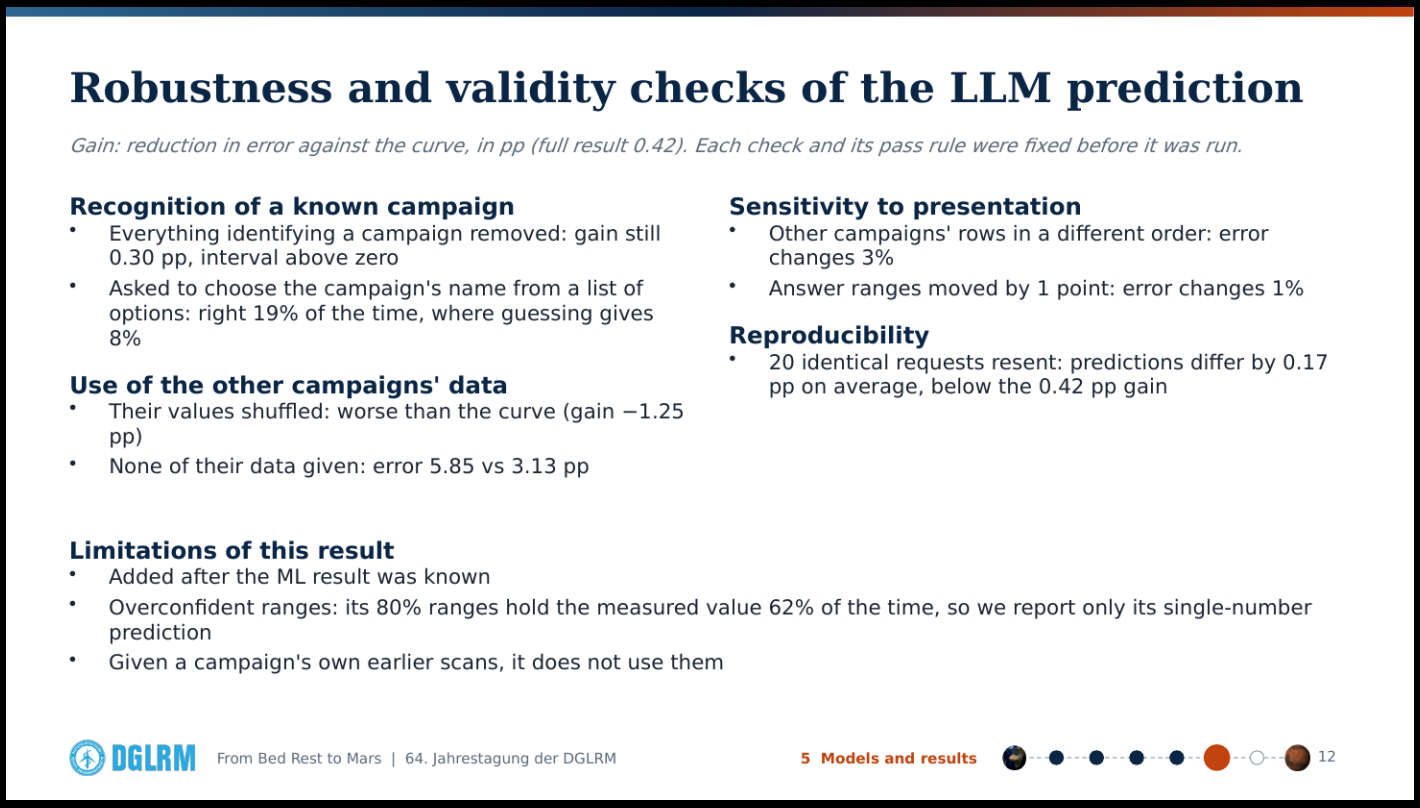
The language model is at the bottom. Its error is 2.71 percentage points, the lowest of all models, and it explains the most variance, 0.43.

Compared with the curve on the same rows and scored the same way, the error falls from 3.13 to 2.71, a gain of 0.42 points. The 95 percent confidence interval runs from 0.13 to 0.73, so even the most pessimistic estimate is an improvement. It is closer than the curve in 20 of the 32 campaigns. No other method has a gain whose interval stays above zero.

If asked about a target: we had set a 15 percent improvement target. The error improvement is 13.3 percent and the probabilistic score improvement 17.7 percent.

Slide 12 · Robustness and validity checks of the LLM prediction

About a minute and a quarter



We wrote down each check and its pass rule, and committed them to the repository, before running it.

Language models have read many papers, so the model might recognise a published campaign and recall its result. With every identifying detail removed, the gain is 0.30 points and its interval stays above zero. When we asked the model to choose the campaign's name from a list of options, it was right 19 percent of the time, where guessing gives 8 percent.

If asked: of the three campaigns behind most of the gain, it named only one.

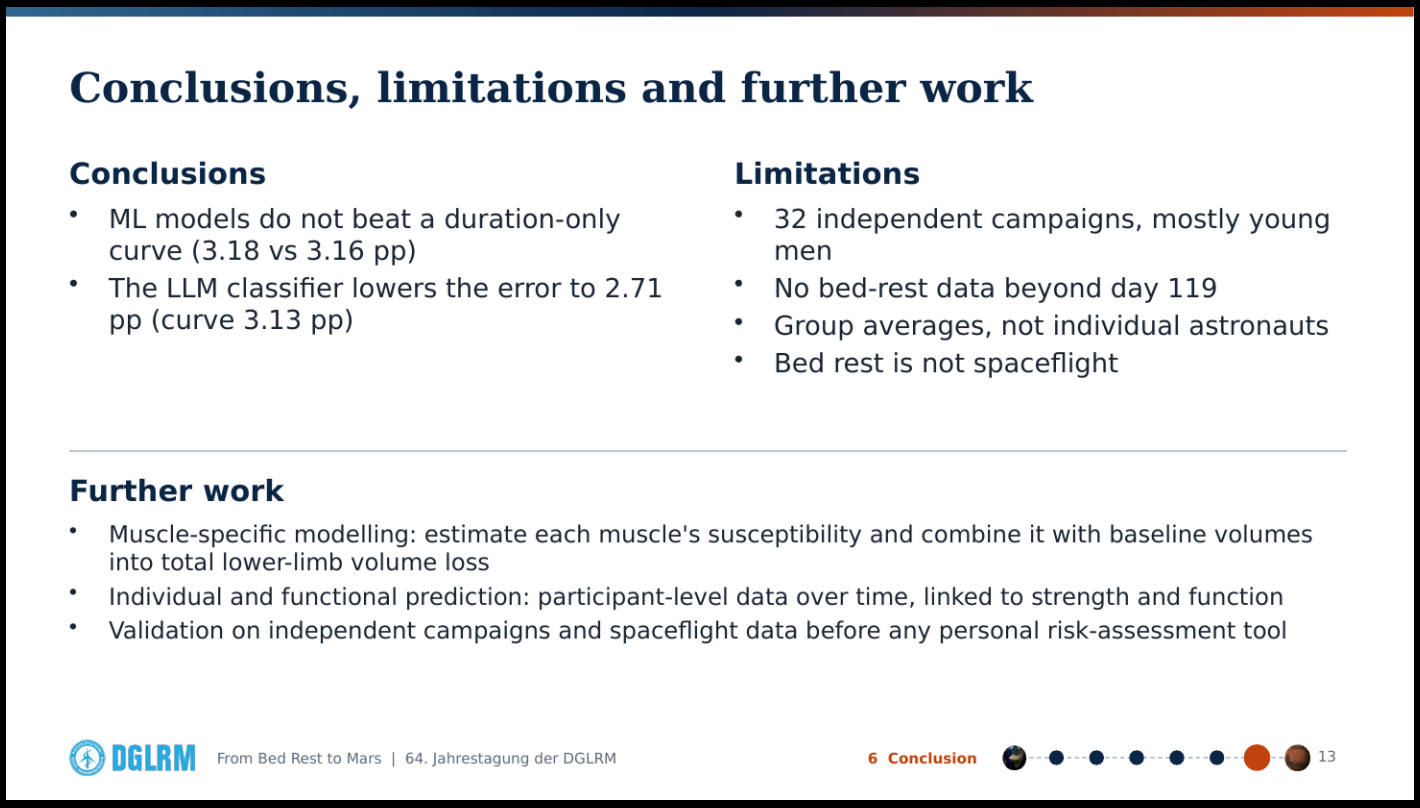
When we shuffle the other campaigns' values, the gain turns into a loss of 1.25 points, and with no reference data the error rises to 5.85. The gain comes from reading those data.

Reordering the rows or shifting the range edges changes the error by 1 to 3 percent. Repeating 20 requests moves the predictions by 0.17 points on average, well under the 0.42 gain.

Three limitations: we added this model after the ML result; its 80 percent ranges contain the truth only 62 percent of the time, so we quote only point predictions; and when given a campaign's own earlier scans, it does not use them.

Slide 13 · Conclusions, limitations and further work

About thirty seconds



With 32 independent campaigns, standard ML models and TabPFN do not beat a curve that only knows the number of days. An LLM-based probabilistic classifier that reads each campaign's description and the other campaigns' data reduces the error from 3.13 to 2.71 points. The main limitation is the small sample of 32 campaigns, mostly young men, with no bed-rest data beyond day 119. Every row is a group average, so the model predicts the average loss of a group of volunteers. It cannot yet predict the risk for an individual astronaut.

Further work has three steps. First, estimate how susceptible each muscle is relative to a reference muscle, and combine those coefficients with baseline muscle volumes to estimate total lower-limb volume loss; the dataset has 30 group time points where three or more muscles were scanned together, across 14 campaigns, so this can be tested. Second, move from group averages to individuals with participant-level data over time, and link muscle loss to strength and function. Third, validate on independent campaigns and on spaceflight data before building any personal risk-assessment tool.

Slide 14 · Thank you for listening

Closing



Close (Milad). Thank the audience and invite questions. The backup slides that follow answer the questions we expect: part 1 on slides 17 and 18, data and search on 15 and 16, part 2 on 19 to 21.

Slide 15 · Backup: dataset coverage and limitations

Backup: only if asked

Backup: dataset coverage and limitations

Did you miss studies?

- Four sources searched from 2013 on, plus 10 older or open-data studies (sources and queries on the next slide)
- Not covered: Embase (no access)

One campaign dominates the data

- MEDES 90-day: 40% of all rows, a quarter of the modelling data
- Scoring gives each campaign one vote; the re-run without it is still to do

Can MRI, CT, DXA and ultrasound be mixed?

- Each method gets its own term: DXA and ultrasound thickness show 5 to 6 pp less loss than CT; MRI volume is within 0.5 pp
- The MRI-only re-run is still to do

How reliable is the extraction?

- Every value is traced to its paper and page, table or figure
- 674 of 742 rows high confidence; 19 read off figures; no independent second extraction

Backup. Be direct that the three must-show sensitivity analyses (composite-first, without MEDES, MRI only) have not been run yet. Only the weighting check has, and it changed little.

Slide 16 · Backup: literature search sources and query

Backup: only if asked

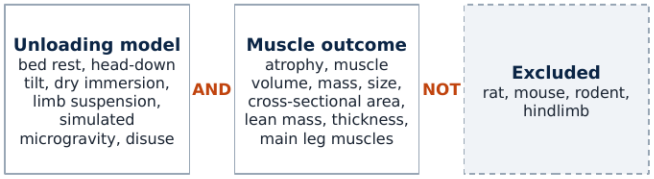
Backup: literature search sources and query

Table 3. Sources (searched 4 September 2026)

Source	Records
PubMed	1,412
Scopus	1,757
Web of Science Core Collection	1,876
NASA Technical Reports Server	686
Earlier studies and NASA open data	10
Total	**5,741**

- Journal databases: publications from 2013 onwards
- Older work enters through studies we held before the search (9) and one NASA open-data campaign
- NASA reports: five short keyword queries, e.g. "bed rest" "muscle volume"
- Not searched: Embase (no institutional access), Cochrane CENTRAL, trial registries, citation chasing

Figure 12. Query: two required blocks



Web of Science string, verbatim

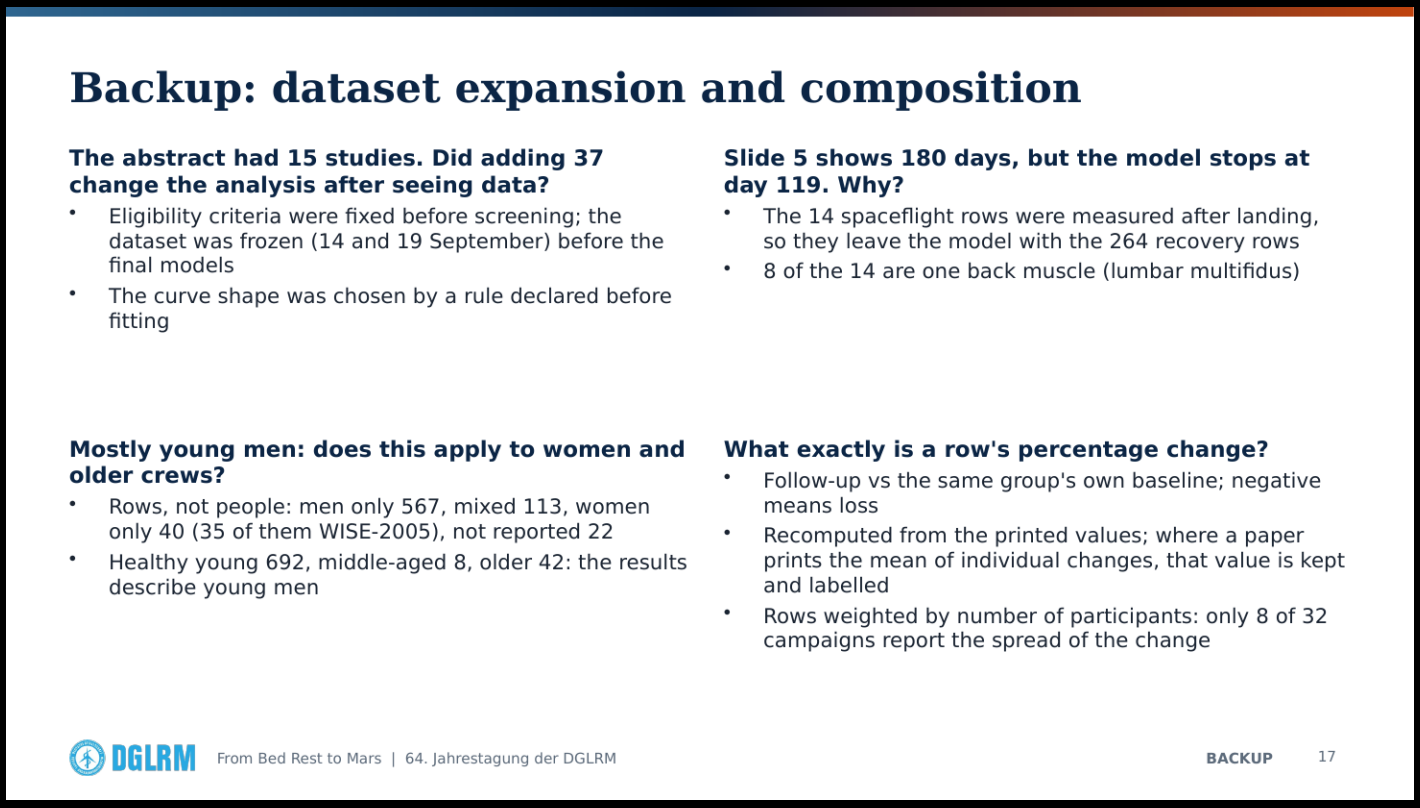
```
TS=(("bed rest" OR bedrest OR "head-down tilt" OR "head down bed rest" OR HDBR
  OR antiorthostatic OR hypokinesia OR hypodynamia OR "dry immersion"
  OR "limb suspension" OR ULLS OR "simulated microgravity"
  OR "microgravity analog*" OR "spaceflight analog*" OR "mechanical unloading"
  OR "muscle unloading" OR disuse)
 AND
 (atroph* OR "muscle volume" OR "muscle mass" OR "muscle size"
  OR "cross-sectional area" OR PCSA OR "lean mass" OR "muscle thickness"
  OR "muscle wasting" OR deconditioning OR soleus OR gastrocnemius
  OR "triceps surae" OR quadriceps OR "vastus lateralis" OR "knee extensor"))
NOT TS=(rat OR rats OR mice OR mouse OR rodent OR hindlimb OR "hind limb")
```

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Backup. All four strings are in the report appendix, copied from docs/literature-review/search_log.md. PubMed uses the same two blocks with MeSH terms and the humans filter; Scopus and Web of Science exclude animal studies in the query. The 2013 limit is the main weakness: only two of ten known modelling papers fall inside the window, and the known-item test without the limit was not run. Embase was not searched for lack of access; it indexes conference abstracts the other databases miss.

Slide 17 · Backup: dataset expansion and composition

Backup: only if asked



Backup, part 1 (Niloufar). Expansion: the eligibility criteria in the report's Table 2 were written before screening; dataset version 1.0 was frozen on 14 September and 1.1 on 19 September, and every result is fitted on 1.1. The rule for the curve: the saturating form is the headline unless another form beats it by at least 4 AIC points. Spaceflight: the dataset keeps the spaceflight rows, flagged, but they are post-flight measurements, so the during-unloading filter removes them. Weighting: inverse-variance weighting would need the spread of the change, which most papers do not print; the check that compares both weightings moved the duration effect by under 1 pp.

Slide 18 · Backup: certainty of the muscle-group curves

Backup: only if asked

Backup: certainty of the muscle-group curves

Is each line fitted to its own muscle?

- No. The day-60 values are fitted per muscle group (named muscles, 25 campaigns); the lines share one time course (τ = 60 days)
- Separate shapes cannot be estimated: six of nine groups rest on four campaigns or fewer

Why a saturating curve?

- Logarithmic, saturating and spline curves fit equally well (within 1.3 AIC points): the loss slows, but the data cannot say along which curve
- By day 119 the curve reaches 86% of its plateau, −17.3% (95% CI −20.9 to −13.7)

How certain are the day-60 values?

- Calf −15.8% (95% CI −18.9 to −12.7); front thigh −10.1% (−13.6 to −6.6); hip rotators −1.9% (−3.9 to 0.2)
- Calf vs shin, the ankle's opposing pair: 5.6 pp more loss (2.1 to 9.1), p = 0.003
- Hip rotators and outer hip: one campaign each

Is the calf result an artefact of how often it is scanned?

- The order holds after adjusting for duration, countermeasure group and imaging method
- Calf: 78 rows from 13 campaigns, the second best-covered group after the front thigh (86 rows, 22 campaigns)

Backup, part 1 (Niloufar). Be precise about slide 8: the coloured dots at day 60 are the model's estimates for each muscle group, with the intervals shown here. The lines are drawn with one shared time course scaled to those values, so they show the shape the model assumes, not a separate fit per muscle. The intervals are 95% cluster-robust, which means they account for several rows coming from the same campaign. The calf-versus-shin contrast is reported on its own because it does not depend on which muscle is the reference.

Slide 19 · Backup: significance and relevance of the LLM gain

Backup: only if asked

Backup: significance and relevance of the LLM gain

Is 0.42 pp worth anything?

- It is 13% lower error than the curve (3.13 to 2.71 pp)
- Small; we present it as a secondary finding

20 of 32 campaigns is close to a coin flip

- Counting wins only: p = 0.22 (sign test)*
- Using the size of each win: mean gain 0.42 pp, 95% CI 0.13 to 0.73

Does it rest on a few campaigns?

- 3 long MRI campaigns (NASA SPRINT, Berlin BBR2-2, MEDES LTBR) give 56% of the gain; without them 0.20 pp, CI −0.01 to 0.43*
- Only SPRINT is recognised, and with campaign details removed the LLM still wins there (3.79 vs 5.92 pp)

Did it read these papers in training? Was it added after the fact?

- No test can rule out training exposure; hiding campaign details and the naming test are the strongest checks available
- Added after the ML result, but every check and target was committed before the first answer

** Our calculation from the saved LLM predictions; not in the report.*

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Backup. The two starred numbers are our own calculation from results/forecast_predictions.csv, not in the report. Without the three campaigns that carry most of the gain, the interval touches zero; say so if asked.

Slide 20 · Backup: ML results and fairness of the comparison

Backup: only if asked

Backup: ML results and fairness of the comparison

Did the ML models get a fair chance?

- Tuned inside each training fold, grouped by campaign; TabPFN needs no tuning
- All models saw the same inputs, folds and scoring

Why not add age, sex or countermeasure type?

- 32 campaigns leave room for about 3 campaign-level inputs (about 10 campaigns per input)
- Age and sex are fixed within a campaign (271 of the 346 modelling rows men-only), so they act as a campaign label; 100 countermeasure rows spread over 9 types

Is the duration curve a straw man?

- Ridge regression uses days, muscle and every other input: 3.28 vs 3.16 pp
- Knowing the muscle raises R^2 but not the error on a new campaign

Would a nearest-neighbour method match the LLM?

- Not tested. If the gain comes from picking similar rows, this is the next control to run

Backup. The honest gap here is the nearest-neighbour control: we cannot yet say whether a simple similarity method would match the LLM. The 271 men-only rows and 100 countermeasure rows are our count on the 346 modelling rows; the report gives the same point on the 425 lower-limb rows before duplicate composites were removed (347 men-only, 126 countermeasure).

Slide 21 · Backup: applicability to Mars mission planning

Backup: only if asked

Backup: applicability to Mars mission planning

A Mars transit is about 180 days; the data stop at 119

- Only one campaign reaches 119 days
- A 180-day estimate is planned from the duration curve only, with an uncertainty range, labelled as an assumption

Bed rest is not spaceflight

- Only 14 spaceflight rows, 8 of them one back muscle: too few to test transfer

Which countermeasure works? What about individuals?

- Countermeasure groups lose 3.7 pp less (95% CI 1.5 to 5.9), all types pooled
- Every row is a group average; nothing about individual risk

Can anyone reproduce an LLM result?

- Fixed model version; all 430 answers saved, so every number rebuilds without the model
- Identical requests differ by 0.17 pp on average

Backup. The supported planning statements, within 5 to 119 days and for groups: most loss happens in the first weeks, calf muscles need protecting first, and countermeasure groups lose less.